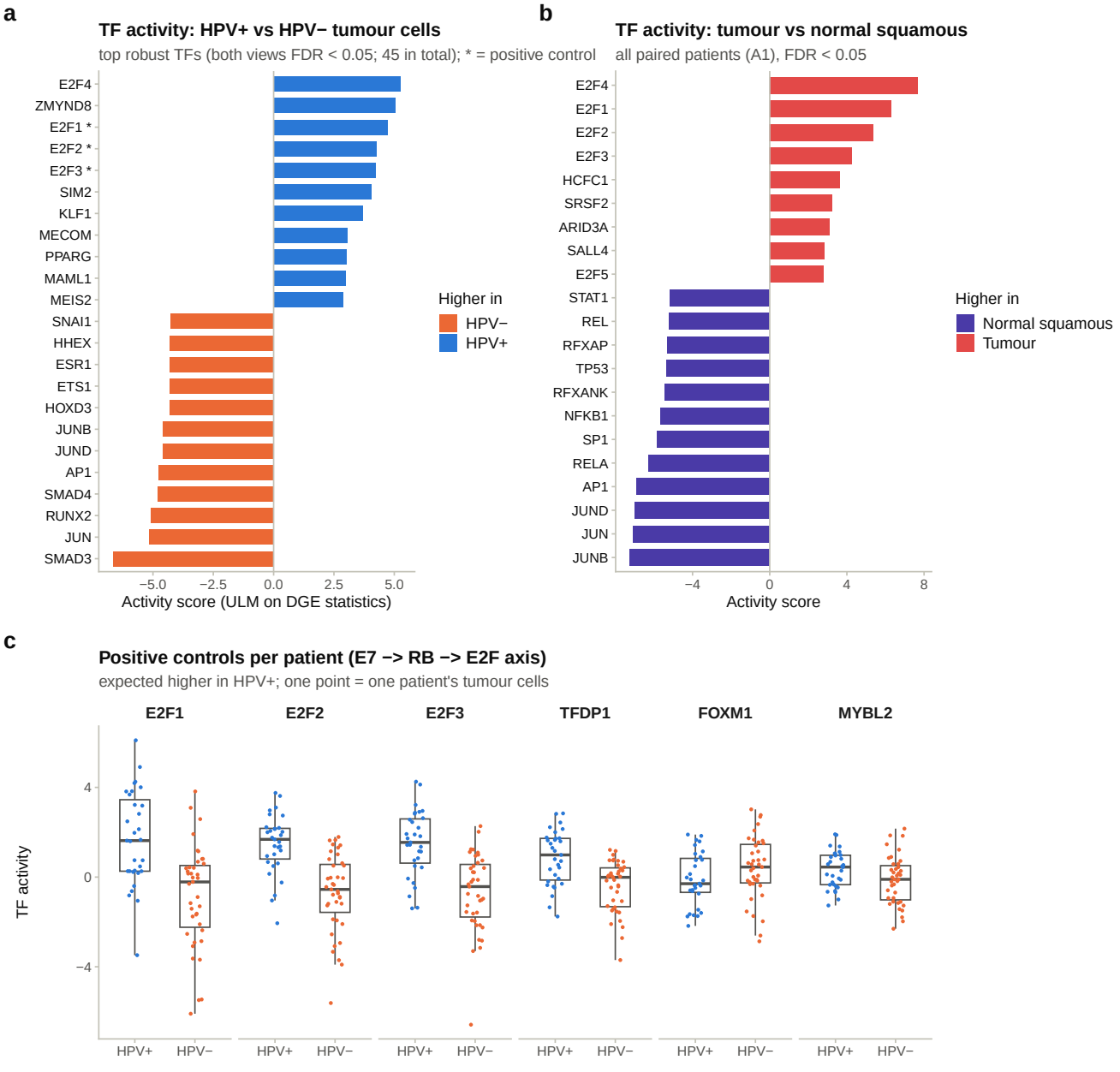
a
TF activity: HPV+ vs HPV− tumour cells
top robust TFs (both views FDR < 0.05; 45 in total); * = positive control
E2F4
ZMYND8
E2F1 *
E2F2 *
E2F3 *
SIM2
KLF1
MECOM
PPARG
MAML1
MEIS2
SNAI1
HHEX
ESR1
ETS1
HOXD3
JUNB
JUND
AP1
SMAD4
RUNX2
JUN
SMAD3
−5.0
−2.5
0.0
2.5
5.0
Activity score (ULM on DGE statistics)
Higher in
HPV−
HPV+
b
TF activity: tumour vs normal squamous
all paired patients (A1), FDR < 0.05
E2F4
E2F1
E2F2
E2F3
HCFC1
SRSF2
ARID3A
SALL4
E2F5
STAT1
REL
RFXAP
TP53
RFXANK
NFKB1
SP1
RELA
AP1
JUND
JUN
JUNB
−4
0
4
8
Activity score
Higher in
Normal squamous
Tumour
c
Positive controls per patient (E7 −> RB −> E2F axis)
expected higher in HPV+; one point = one patient's tumour cells
E2F1
E2F2
E2F3
TFDP1
FOXM1
MYBL2
TF activity
4
0
−4
HPV+
HPV−